

Math 2551 Worksheet: Review for Exam 3

1. Set up an iterated integral in spherical coordinates for $\iiint_E z^2 dV$ where E is the region between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 25$ and inside $z = -\sqrt{\frac{1}{3}(x^2 + y^2)}$.
2. Set up an integral that computes the volume of the solid which is bounded above by the cylinder $z = 4 - x^2$, on the sides by the cylinder $x^2 + y^2 = 4$, and below by the xy -plane using
 - (a) Cartesian coordinates
 - (b) cylindrical coordinates

Which integral would you rather evaluate and why?

3. Find an integral that computes the mass of the wire which lies along the curve $y^2 = x^3$ from $(0, 0)$ to $(1, -1)$ and has density function $\rho(x, y) = 2xy^2$.
4. Show that the field $\mathbf{F} = 2x\mathbf{i} - y^2\mathbf{j} - \frac{4}{1+z^2}\mathbf{k}$ is conservative, find a potential function, and use it to compute the integral

$$\int_C 2x dx - y^2 dy - \frac{4}{1+z^2} dz$$

where C is any path from $(0, 0, 0)$ to $(3, 3, 1)$.

5. Compute $\int_C (6y + x) dx + (y + 2x) dy$ using any method, where C is the circle $(x - 2)^2 + (y - 3)^2 = 4$.
6. Find the flux of the field $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + \mathbf{k}$ through the portion of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant in the direction away from the origin.
7. Use Stokes' theorem to show that the circulation of the field $\mathbf{F} = \langle 2x, 2y, 2z \rangle$ around the boundary curve C of **any** smooth orientable surface S in $\mathbb{R}^3$ is 0.
8. Find the outward flux of $\mathbf{F} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/\sqrt{x^2 + y^2 + z^2}$ through the boundary S of the "thick sphere" D given by the points satisfying $1 \leq x^2 + y^2 + z^2 \leq 4$.